

In [8]:

```
1 import pandas as pd
2 df = pd.read_csv(r"C:\Users\SANDEEP MEDIDA\Desktop\data-viz-class-material\WHR2023.csv")
3
4 print("Original columns:", df.columns.tolist())
5
6 # Select correct columns explicitly
7 df_clean = df[[
8     'Country name',
9     'Ladder score',
10    'Logged GDP per capita',
11    'Social support',
12    'Healthy life expectancy',
13    'Freedom to make life choices',
14    'Generosity',
15    'Perceptions of corruption'
16 ]].copy()
17
18 # Rename properly
19 df_clean.columns = [
20     'Country',
21     'Happiness_Score',
22     'GDP',
23     'Social_Support',
24     'Life_Expectancy',
25     'Freedom',
26     'Generosity',
27     'Corruption'
28 ]
29
30 print(df_clean.head())
```

Original columns: ['Country name', 'Ladder score', 'Standard error of ladder score', 'upperwhisker', 'lowerwhisker', 'Logged GDP per capita', 'Social support', 'Healthy life expectancy', 'Freedom to make life choices', 'Generosity', 'Perceptions of corruption', 'Ladder score in Dystopia', 'Explained by: Log GDP per capita', 'Explained by: Social support', 'Explained by: Healthy life expectancy', 'Explained by: Freedom to make life choices', 'Explained by: Generosity', 'Explained by: Perceptions of corruption', 'Dystopia + residual']

Country	Happiness_Score	GDP	Social_Support	Life_Expectancy	\ 0	Finland
7.804	10.792	0.969	71.150			
1	Denmark	7.586	10.962	0.954	71.250	
2	Iceland	7.530	10.896	0.983	72.050	
3	Israel	7.473	10.639	0.943	72.697	
4	Netherlands	7.403	10.942	0.930	71.550	

	Freedom	Generosity	Corruption	0
0.961	-0.019	0.182		
1	0.934	0.134	0.196	
2	0.936	0.211	0.668	

In

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3      0.809      -0.023      0.708
4      0.887      0.213      0.379
[10]: 1 df = pd.read_csv(r"C:\Users\SANDEEP MEDIDA\Desktop\data-viz-class-material\WHR2023.csv" 2
      2 print("Original columns:", df.columns.tolist())
      3
      4      df_subset = df.iloc[:, :9] # Select only the first 9 columns
      5      df_subset.columns =
      6      ['Country', 'Region', 'Happiness_Score', 'GDP', 'Social_Support',
      7      'Life_Expectancy', 'Freedom', 'Generosity', 'Corruption']
      8 print(f"Dataset: {len(df_subset)} countries, {len(df_subset.columns)} columns") 9
      9 print(df_subset.head())
```

Original columns: ['Country name', 'Ladder score', 'Standard error of ladder score', 'upperwhisker', 'lowerwhisker', 'Logged GDP per capita', 'Social support', 'Healthy life expectancy', 'Freedom to make life choices', 'Generosity', 'Perceptions of corruption', 'Ladder score in Dystopia', 'Explained by: Log GDP per capita', 'Explained by: Social support', 'Explained by: Healthy life expectancy', 'Explained by: Freedom to make life choices', 'Explained by: Generosity', 'Explained by: Perceptions of corruption', 'Dystopia + residual']
Dataset: 137 countries, 9 columns

	Country	Region	Happiness_Score	GDP	Social_Support	\ 0
	Finland	7.804	0.036	7.875	7.733	
1	Denmark	7.586	0.041	7.667	7.506	
2	Iceland	7.530	0.049	7.625	7.434	
3	Israel	7.473	0.032	7.535	7.411	
4	Netherlands	7.403	0.029	7.460	7.346	

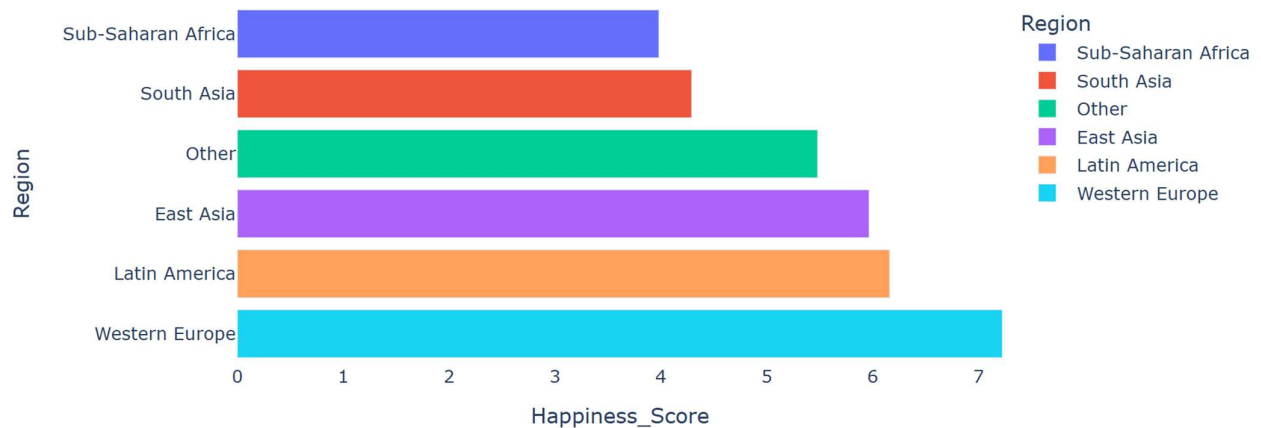
	Life_Expectancy	Freedom	Generosity	Corruption	0
	10.792	0.969	71.150	0.961	
1	10.962	0.954	71.250	0.934	
2	10.896	0.983	72.050	0.936	
3	10.639	0.943	72.697	0.809	
4	10.942	0.930	71.550	0.887	

In

```
1 region_map = {
2     'Finland': 'Western Europe',
3     'Denmark': 'Western Europe',
4     'Iceland': 'Western Europe',
5     'Netherlands': 'Western Europe',
6     'Sweden': 'Western Europe',
7     'Norway': 'Western Europe',
8     'Germany': 'Western Europe',
9     'France': 'Western Europe',
10    'Spain': 'Western Europe',
11
12    'India': 'South Asia',
13    'Pakistan': 'South Asia',
14    'Bangladesh': 'South Asia',
15
16    'China': 'East Asia',
17    'Japan': 'East Asia',
18    'South Korea': 'East Asia',
19
20    'Nigeria': 'Sub-Saharan Africa',
21    'Kenya': 'Sub-Saharan Africa',
22    'Ethiopia': 'Sub-Saharan Africa',
23    'Congo (Kinshasa)': 'Sub-Saharan Africa',
24    'Sierra Leone': 'Sub-Saharan Africa',
25
26    'Brazil': 'Latin America',
27    'Argentina': 'Latin America',
28    'Mexico': 'Latin America'
29 }
30
31 df_clean['Region'] = df_clean['Country'].map(region_map)
32 df_clean = df_clean.dropna(subset=['Region'])
```

In [11]:

Western Europe Shows the Highest Happiness Levels Among Selected Regions



In

```
1 import pandas as pd
2 import plotly.express as px # <-- add this
3
4 region_avg = df_clean.groupby('Region')['Happiness_Score'].mean().reset_index()
5 region_avg = region_avg.sort_values('Happiness_Score')
6
7 fig1 = px.bar(
8     region_avg,
9     x='Happiness_Score',
10    y='Region',
11    orientation='h'
12 )
13
14 fig1.update_xaxes(range=[0, region_avg['Happiness_Score'].max()])
15
16 fig1.update_layout(
17     title="Western Europe Shows the Highest Happiness Levels Among Selected Regions" ,
18     plot_bgcolor='white'
19 )
20
21 fig1.show()
22
```

[20]:



In

[17]:

```
1 import plotly.express as px
2
3 # Top and Bottom
4 top8 = df_clean.nlargest(8, 'Happiness_Score').copy()
5 top8['Group'] = 'Top 8'
6
7 bottom8 = df_clean.nsmallest(8, 'Happiness_Score').copy()
8 bottom8['Group'] = 'Bottom 8'
9
10 combined = pd.concat([bottom8, top8])
11 combined = combined.sort_values('Happiness_Score')
12
13 # Global average
14 global_avg = df_clean['Happiness_Score'].mean()
15
16 # Plot
17 fig = px.bar(
18     combined,
19     x='Happiness_Score',
20     y='Country',
21     orientation='h',
22     color='Group',
23     color_discrete_map={'Top 8': 'green', 'Bottom 8': 'red'}
24 )
25
26 fig.add_vline(x=global_avg, line_dash="dash")
27
28 fig.update_xaxes(range=[0, combined['Happiness_Score'].max()])
29
30 fig.update_layout(
31     title="Top Countries Are Nearly Twice as Happy as the Lowest Ranked Ones",
32     plot_bgcolor='white'
33 )
34
35 fig.show()
36
```

Western Europe Leads in Both Economic Strength and Personal Freedom Compared to Other Re

